

FAIRHOPE, AL, USA

WMO: 720592

Lat: 30.460N

Lon: 87.877W

Elev: 28

StdP: 100.99

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00186

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	0.0	-9.8	1.6	2.6	-7.1	2.1	3.9	9.4	19.2	8.6	16.7	1.8	0	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	34.1	25.9	32.8	25.6	32.3	25.5	27.7	30.8	27.4	30.1	27.0	29.8	2.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.1	28.8	26.4	21.9	28.5	26.2	21.7	28.4	88.1	30.4	86.1	29.9	84.4	29.9	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.1	7.1	5.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	19.7	10.7	13.2	15.7	19.2	22.8	26.7	27.3	27.1	25.4	20.4	14.4	12.6
	DBStd	7.01	5.51	5.12	4.43	3.15	2.79	2.10	1.72	1.83	2.03	3.86	4.54	4.94
	HDD10.0	134	59	25	8	0	0	0	0	0	0	0	12	30
	HDD18.3	872	242	154	103	26	4	0	0	0	0	24	132	187
	CDD10.0	3661	81	114	185	277	397	500	536	531	463	323	145	109
	CDD18.3	1357	6	10	22	53	143	250	277	272	213	89	14	8
	CDH23.3	12533	8	20	72	362	1203	2633	2916	2722	1757	758	72	10
	CDH26.7	4490	0	0	4	38	298	1057	1168	1098	639	185	3	0
(14) Wind	WSAvg	2.2	2.8	2.9	2.9	2.7	2.2	1.8	1.5	1.5	1.6	2.1	2.2	2.7
(15) Precipitation	PrecAvg	1581	129	118	138	119	106	147	187	161	150	92	111	123
	PrecMax	2197	402	249	295	433	338	354	398	253	580	319	297	326
	PrecMin	1050	30	28	14	20	17	34	58	61	40	1	30	49
	PrecStd	294	82	53	77	87	70	82	73	61	106	79	75	55
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.7	25.2	27.3	28.9	32.4	35.9	35.9	36.0	33.8	31.3	27.3	24.9
		MCWB	20.8	20.5	19.6	22.3	23.7	25.2	26.4	26.0	25.1	24.3	21.3	20.3
	2%	DB	22.6	23.0	25.2	27.6	30.9	33.8	33.8	34.0	32.4	30.1	25.2	22.9
		MCWB	20.3	20.3	20.2	21.1	23.5	25.6	26.0	26.4	25.0	23.3	20.5	20.4
	5%	DB	21.3	22.3	23.7	27.1	29.8	32.5	32.6	32.5	31.3	28.7	23.7	22.2
		MCWB	19.9	19.7	19.7	20.8	23.2	25.2	26.0	26.0	24.7	23.4	19.6	20.3
	10%	DB	20.0	21.1	22.4	25.9	27.9	31.3	31.3	31.3	30.2	27.5	22.4	21.0
		MCWB	18.6	19.2	19.4	20.4	22.8	25.0	25.9	25.7	24.7	22.1	18.9	19.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	22.6	22.4	24.8	25.8	27.9	28.2	28.5	27.4	27.4	23.5	22.5
		MCDB	23.5	23.8	24.3	26.9	27.3	31.0	31.3	33.0	30.5	29.1	25.2	23.3
	2%	WB	21.1	21.4	21.8	23.5	25.1	27.3	27.6	27.8	26.7	26.0	22.4	21.4
		MCDB	22.1	22.5	23.6	25.9	27.7	30.0	30.5	31.1	29.6	28.0	24.3	22.1
	5%	WB	20.2	20.7	21.1	22.9	24.5	26.8	27.2	27.4	26.2	24.8	21.2	20.7
		MCDB	21.0	21.5	22.5	25.1	27.8	29.7	30.2	30.1	29.0	27.2	22.6	21.4
	10%	WB	18.6	19.4	20.3	22.1	23.9	26.3	26.8	27.0	25.6	23.5	19.8	19.6
		MCDB	20.0	20.4	21.9	24.6	27.4	29.3	29.8	29.7	28.6	26.3	21.6	20.5
(35) Mean Daily Temperature Range	5% DB	MDBR	11.0	10.4	10.7	10.9	10.1	8.5	8.2	8.4	9.4	11.9	12.2	10.1
		MCDBR	9.8	9.8	10.3	12.0	11.5	10.2	9.5	9.7	10.5	12.1	12.1	9.9
		MCWBR	7.6	6.7	6.2	6.5	5.2	3.9	3.9	4.1	4.6	6.1	7.9	7.8
	5% WB	MCDBR	8.2	7.9	8.1	8.5	8.9	7.9	8.0	7.9	8.6	8.3	9.9	8.7
		MCWBR	7.2	6.3	5.9	6.0	4.7	4.1	4.1	4.3	4.6	5.3	7.8	8.0
(40) Clear-Sky Solar Irradiance	taub		0.332	0.349	0.373	0.403	0.438	0.463	0.509	0.498	0.444	0.383	0.355	0.341
	taud		2.547	2.503	2.430	2.347	2.285	2.267	2.160	2.185	2.303	2.458	2.477	2.531
	Ebn at Noon		902	915	912	891	857	831	791	797	830	865	865	871
	Edh at Noon		86	98	112	126	135	137	152	146	126	101	91	83
(44) All-Sky Solar Radiation	RadAvg		3.07	3.64	4.93	5.98	6.63	6.29	5.92	5.54	5.21	4.57	3.50	2.70
	RadStd		0.24	0.50	0.39	0.48	0.34	0.48	0.41	0.41	0.31	0.45	0.30	0.24

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (30 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	+104

Nomenclature: See separate page